

RICHA RANJAN

SOFTWARE ENGINEER

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PROFILE

B.Tech graduate in Computer Science and Business Systems with a strong foundation in Java, JavaScript, MySQL, and core Computer Science fundamentals. Experienced in designing, building, and debugging full-stack applications using clean, maintainable architecture (MVC) and RESTful APIs. Skilled at translating requirements into design documentation and refactoring code to improve performance, reliability, and maintainability. Practical exposure to machine learning (Scikit-learn, Decision Trees, Random Forest) and data-driven problem-solving, with strong command of Git-based version control, CI/CD workflows, and Agile/Scrum practices.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript

Web & Architecture: HTML5, CSS3, React.js, Node.js, Express.js, REST API Design, MVC Architecture

Databases: MySQL, MongoDB, Database Design, Query Optimization, CRUD Operations

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, SDLC

Machine Learning: Scikit-learn, Decision Trees, Random Forest, Data Preprocessing, Feature Engineering

Tools & Practices: Git, GitHub, CI/CD, Postman, VS Code, Agile/Scrum, Requirements Analysis, Design Documentation, Code Optimization & Refactoring, Debugging

PROJECTS

Food Delivery Website

MERN Stack, REST APIs, JWT

- Architected a full-stack food delivery application using MongoDB, Express.js, React.js, and Node.js (MVC pattern) with JWT authentication and role-based access control.
- Designed normalized database schemas and optimized RESTful CRUD endpoints for catalog, cart, and order management.

Crop Recommendation System

Python, Scikit-learn, Machine Learning

- Trained and evaluated Decision Tree and Random Forest models with data preprocessing and feature engineering, achieving **94.5% prediction accuracy** on agricultural datasets.

Spreadsheet Enhancement System

JavaScript, HTML, CSS

- Built Excel-like functionality including dynamic formula evaluation, real-time cell updates, and theme toggling in vanilla JavaScript.
- Optimized DOM manipulation and rendering for large datasets, improving UI speed.

PROFESSIONAL EXPERIENCE

Web Developer Intern — CODESOFT

June 2023 – July 2023

- Built a fully responsive portfolio dashboard using HTML5, CSS3, and JavaScript, achieving **100% UI compatibility** across 15+ screen sizes.
- Cut page load time by **40%** through performance optimization and lazy loading techniques.
- Debugged and resolved **20+ UI/performance bugs** using Git version control, reducing post-release issue reports by **~30%** while collaborating in an Agile environment.

EDUCATION

B.Tech, Computer Science & Business Systems — Asansol Engineering College, West Bengal

2022 – 2026

CGPA: 7.64/10

Senior Secondary Education (Science) — D.A.V Public School, Barora, Dhanbad

2022

66.4%

Secondary Education — D.A.V Public School, Barora, Dhanbad

2020

77.4%

ACHIEVEMENTS & CERTIFICATIONS

- Solved **450+ Data Structures & Algorithms problems** on GeeksforGeeks across arrays, trees, graphs, dynamic programming, and sorting/searching.
- Ranked **8th in Smart India Hackathon (SIH) 2024** — College Round, a national-level innovation and software development competition.
- Completed an Ethical Hacking Workshop at IIT Kharagpur, gaining exposure to cybersecurity fundamentals and network security.